Mic rail (D-067 F): passive, so never above the Pi's 3V3. 2.2 ohm damps the bead/C resonance near 34 kHz (Q 1.6 -> 0.12).

Amp: W12 5 V only (D-067 A). BCLK/LRCLK/DIN are 6 V-tolerant with VDD off; the 100 ohm resistors are edge damping.

SD_MODE is rated to VDD + 0.3 V: 2.2k limits it to ~1.2 mA with the branch off (datasheet remedy). No power-good path (D-067 C).

J2-J5: W27 mic leads, GH 1 3V3, 2 CLK, 3 GND, 4 DATA (PCB-06). REAR on J5-2/J5-4, FRONT on J5-3/J5-5 (D-067 D).

PDM clock: one series resistor per lead, source termination, edge <= 13 ns at the mic; tune on the bench).

SDATA tri-states outside its 20 bits: 100k pull-down per lane (ADAU7002 Fig. 11).

GAIN_SLOT: both open = 9 dB (default). R13 to GND: OR 12 dB, 100k 15 dB. R14 to VDD: OR 6 dB, 100k 3 dB. Fit one after the BO-007 bench listen (D-034 capped gain).

Unfitted 5V-present divider (5.0 V -> 3.0 V) on J5-1 pin 10, for a later CA-06 change (D-067 C).

H1-H3: M2 SMT spacers (8 mm) reflowed on the top face, no net. Board-local (3.5, 30.0), (3.5, 8.0), (18.0, 19.0) (D-067 E).

Spec: docs/03-build/04-pcbs/pcb-05-audio-front-end/README.md
2 x ADAU7002 (PDM -> I2S), MAX98357A (I2S -> class D), mic 3.3 V filter (D-067)

**Makad**

Sheet: /
File: pcb-05-audio-front-end.kicad_sch

**Title: PCB-05 audio front end**

| Size: A3 | Date: 2026-10-27 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.99.0-5923-gde08734017 | | Id: 1/1 |